Design & Analysis of a Semi-Active Suspension System Using Magnetorheological (MR) Dampers

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AIM

To design and analyze a semi-active vehicle suspension system using a Magnetorheological (MR) damper and evaluate its ability to improve ride comfort and handling stability compared with a conventional passive suspension system.

ABSTRACT

Conventional passive suspension systems rely on a fixed spring-damper configuration that represents a single compromise between ride comfort and vehicle handling. Because this configuration cannot adapt to changing road conditions, driving speeds, or load, it inevitably under-performs in at least one of these areas at any given time. This project proposes and analyses a semi-active suspension system that uses a Magnetorheological (MR) damper as its core actuator. MR dampers use a fluid containing micron-sized iron particles whose apparent viscosity changes almost instantaneously when exposed to a magnetic field, allowing the damping force to be varied electronically without any mechanical valves. A two-degree-of-freedom quarter-car model is used to represent the sprung mass (chassis) and unsprung mass (wheel assembly), and a closed-loop control strategy (Skyhook / PID) continuously adjusts the current supplied to the damper based on real-time accelerometer feedback. The system is evaluated in simulation against a standard passive suspension baseline, using body acceleration (RMS), suspension travel, and tyre deflection as the primary performance metrics under bump and random road inputs. The project aims to demonstrate that a semi-active MR-damper suspension can approach the ride comfort and stability of a fully active suspension system while keeping power consumption, mechanical complexity, and cost close to that of a conventional passive system, making it a practical and scalable solution for passenger, off-road, and rail applications.

OBJECTIVES

- Develop a 2-DOF quarter-car model incorporating an MR damper as the semi-active control element.
- Develop a real-time control strategy (Skyhook / PID) to vary damping force.
- Simulate and compare passive vs. semi-active response to standard road inputs.
- Quantify improvement in ride comfort (RMS acceleration) and handling stability.

METHODOLOGY OVERVIEW

Literature Review	Survey passive, semi-active, and active suspension research and existing MR damper models.
Mathematical Modelling	Derive the 2-DOF quarter-car equations of motion for sprung and unsprung mass.
Controller Design	Develop and tune a Skyhook / PID control strategy for the damping force.
Simulation	Model the system in MATLAB/Simulink under bump and random road inputs.
Result Comparison	Evaluate RMS body acceleration, suspension travel, and tyre deflection.
Prototype Planning	Outline the hardware needed for a physical proof-of-concept test rig.

EXPECTED OUTCOMES

Based on benchmarks drawn from published MR-damper quarter-car studies, the project targets approximately a 35% reduction in RMS body acceleration, a 22% reduction in suspension travel, and a 28% improvement in tyre-road contact consistency relative to a passive baseline, confirming the practical viability of MR-damper-based semi-active suspension for real-world vehicle applications.